

LAB 2 — TRADITIONAL AI, GENERATIVE AI AND COMBINED AI IN AVIATION

ACTIVITY 1 — BUILD THE CLASSIFICATION RULE

Traditional AI is suitable when an aviation organisation needs to:

Predict, classify, detect, score, optimise or recommend something using existing data.

Examples:

- Predict passenger demand
- Predict flight delays
- Detect unusual payment patterns
- Forecast airport passenger traffic
- Recommend an operational route

Generative AI is suitable when an aviation organisation needs to:

Create new content such as text, images, summaries, explanations or personalised messages.

Examples:

- Draft a flight-delay message
- Create advertisements
- Summarise a passenger-satisfaction report
- Write a job description
- Create a promotional image

Both may be required when:

Traditional AI first analyses or predicts something, and Generative AI then communicates or explains the result.

Example:

Traditional AI predicts that a flight is likely to be delayed → Generative AI prepares a personalised passenger notification.

These rules follow the lab's stated distinction: Traditional AI predicts/classifies/detects/optimises/recommends, while Generative AI creates content.

ACTIVITY 2 & 3 — CLASSIFICATION OF ALL 15 SITUATIONS

No	Aviation Situation	Classification	Reason
1	Predict passenger demand for	Traditional AI	It uses historical and operational data to

	next month's flights		forecast future demand.
2	Draft a message informing passengers about a flight delay	Generative AI	It creates new written communication for passengers.
3	Detect unusual booking/payment patterns for further review	Traditional AI	It detects unusual patterns in existing data.
4	Create three advertisements for a new airline route	Generative AI	It generates new advertising content.
5	Recommend a fuel-efficient route using operational data	Traditional AI	It analyses data and recommends an operational option.
6	Make the final decision whether an aircraft is airworthy	Primarily Human Judgement	Airworthiness is a high-stakes business-like decision requiring authorised human responsibility.
7	Predict which flights are likely to experience delays	Traditional AI	It predicts delays using operational data.
8	Write personalised notifications for affected passengers	Generative AI	It generates personalised passenger communication.
9	Predict a flight delay and prepare a passenger notification	Both	Traditional AI predicts the delay and Generative AI prepares the message.
10	Make the final decision during an aviation emergency	Primarily Human Judgement	Emergency decisions involve significant safety responsibility.
11	Summarise an uploaded passenger-satisfaction report	Generative AI	It generates a concise summary from the provided report.
12	Create a promotional image for a new destination campaign	Generative AI	It creates new visual content.
13	Forecast passenger traffic at an airport	Traditional AI	It predicts future passenger traffic using data.
14	Write a job description for cabin-crew recruitment	Generative AI	It generates recruitment content.
15	Make the final decision in a serious safety investigation	Primarily Human Judgement	The final outcome carries major safety and business-like responsibility.

The lab's recommended classifications confirm these distinctions, including Traditional AI for demand/delay prediction, Generative AI for drafting/creation, Both for delay prediction plus notification, and authorised human judgement for high-stakes decisions.

COMPARISON RECORD

You can fill your comparison table like this:

Situation	My Answer	ChatGPT Answer	Same/ Different	Final Decision
1	Traditional AI	Traditional AI	Same	Traditional AI

2	Generative AI	Generative AI	Same	Generative AI
3	Traditional AI	Traditional AI	Same	Traditional AI
4	Generative AI	Generative AI	Same	Generative AI
5	Traditional AI	Traditional AI	Same	Traditional AI
6	Human Judgement	Human Judgement	Same	Human Judgement
7	Traditional AI	Traditional AI	Same	Traditional AI
8	Generative AI	Generative AI	Same	Generative AI
9	Both	Both	Same	Both
10	Human Judgement	Human Judgement	Same	Human Judgement
11	Generative AI	Generative AI	Same	Generative AI
12	Generative AI	Generative AI	Same	Generative AI
13	Traditional AI	Traditional AI	Same	Traditional AI
14	Generative AI	Generative AI	Same	Generative AI
15	Human Judgement	Human Judgement	Same	Human Judgement

Reflection on comparison

1. In which situations did your answer differ from ChatGPT?

My answers did not differ significantly from ChatGPT's answers as the classification rules were straightforward.

2. Was ChatGPT's explanation convincing?

Yes. The explanations were straightforward as they connected each task with the type of output produced.

3. Did you change any answer?

No significant changes were needed.

4. Did ChatGPT attempt to automate any high-stakes aviation decision?

No. High-stakes decisions such as airworthiness, emergencies and serious safety investigations were kept under authorised human judgement.

5. Which decisions should remain under authorised human control?

Safety-critical, regulatory, legal, ethical and other high-responsibility decisions should remain under authorised human control.

ACTIVITY 5 — SELECTED CASE

I'll select CASE A — AIRLINE CUSTOMER SERVICE as it clearly demonstrates how Traditional AI and Generative AI can work together.

The case is: an airline wants to predict which flights are likely to be delayed and communicate with affected passengers.

1. Traditional AI Task

Predict which flights are likely to experience delays using available operational data.

2. Generative AI Task

Prepare personalised passenger messages about the verified delay information.

3. Combined Workflow

Operational data → Traditional AI predicts delay → Prediction reviewed → Generative AI drafts passenger message → Human verifies → Message sent

4. Information Required

- Flight status
- Scheduled departure/arrival information
- Available operational data
- Verified delay information
- Passenger communication requirements

5. Human Responsibility

An authorised airline employee should verify the prediction and the generated communication before key passenger communication is issued.

6. Possible Risk

Incorrect prediction or inaccurate generated details could confuse passengers or cause them to make incorrect travel decisions.

ACTIVITY 6 — ORIGINAL COMBINED AI WORKFLOW

The lab asks the workflow to contain operational input, Traditional AI, its output, Generative AI, its output, human review, final action and a risk.

CASE A WORKFLOW

1. Business/Operational Input

Flight and available operational data

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2. Traditional AI Task

Analyse the data and predict whether a flight is likely to be delayed

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3. Traditional AI Output

Predicted delay / likelihood of delay

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4. Generative AI Task

Prepare a straightforward and personalised passenger notification based only on verified details



5. Generative AI Output

Draft passenger-delay message



6. Human Review

Authorised airline professional checks the prediction and message for accuracy



7. Final Business Action

Approved passenger communication is sent through the appropriate airline channel



8. Major Risk

Incorrect AI prediction or inaccurate generated content could provide passengers with misleading details.

ACTIVITY 7 — REVIEW CHATGPT'S WORKFLOW

Checklist

- ☒ The aviation problem is correctly understood.
- ☒ Traditional AI is used for prediction.
- ☒ Generative AI creates the passenger message.
- ☒ The workflow follows a logical sequence.
- ☒ Human review is included.
- ☒ Safety-critical decisions are not transferred completely to AI.
- ☒ No unsupported airline policy is invented.
- ☒ No passenger personal details is invented.
- ☒ No safety procedure is invented.
- ☒ The identified risk is relevant.

My improvement:

Human verification should take place before any passenger communication is sent. The Generative AI should only use verified details and should not independently decide what passengers are entitled to receive.

This follows the lab's requirement that human review remain part of the workflow and that unsupported policies or passenger details should not be invented.

ACTIVITY 8 — IMPROVED WORKFLOW

Improved Case A Workflow

Flight & operational data



Traditional AI

Predicts likelihood of delay



AI Prediction

Possible flight delay identified



Human/authorised business-like verifies relevant details



Generative AI

Creates a passenger communication draft



Human reviews the generated message



Approved message



Passenger communication

Clear division of responsibility

- **Traditional AI:** Prediction
 - **Generative AI:** Communication draft
 - **Human professional:** Verification, approval and responsibility
 - **Final action:** Taken only after appropriate human review
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ACTIVITY 9 — 2 × 2 MATRIX

For Case A — Airline Customer Service:

Question

Response

Is the task mainly creative/drafting or factual/decision-based?

Both — prediction is factual/data-based while passenger communication involves drafting.

Is the output low-stakes or high-stakes?

Moderate to high impact, because passengers may rely on the information.

What could happen if the output is incorrect?

Passengers could receive misleading or confusing information.

Could passengers be affected?

Yes.

Could airline/airport operations be affected?

Yes, through passenger communication and operational coordination.

Question

Response

Could there be a safety or regulatory consequence?

Potentially, depending on the information and situation.

What level of human review is required?

Appropriate human verification before important communication is issued.

Should AI assist or make the final decision?

AI should assist; authorised humans should make final decisions.

Decision Option

AI may assist, but facts must be independently verified.

For high-stakes aviation matters, the lab specifically emphasises that AI should not replace business-like accountability.

ACTIVITY 10 — HALLUCINATION CHALLENGE

The given prompt asks ChatGPT to provide compensation, hotel, meal and refund details without providing the airline's policy, ticket conditions or applicable regulation.

Did ChatGPT invent any information?

It could. If ChatGPT gives a specific compensation amount, hotel entitlement, meal entitlement or refund timeline without an official source, that details would be unsupported.

Which statements require verification?

The following require verification:

- Compensation amount
- Hotel entitlement
- Meal entitlement
- Refund timeline
- Passenger eligibility
- Applicable regulation
- Airline-specific policy
- Ticket conditions

Which official source would you check?

I would check:

The airline's official policy and applicable official aviation/regulatory sources for the passenger's journey.

Improved Prompt

Rewrite the response without inventing any airline policy, compensation amount, passenger entitlement, regulation or refund timeline. Wherever verified details is needed, write [VERIFY OFFICIAL POLICY/REGULATION].

Learning Point

Fluency does not mean accuracy.

A response can sound business-like while still containing unsupported details. The lab specifically highlights this principle as “Fluency ≠ Accuracy.”

ACTIVITY 11 — FINAL AVIATION WORKFLOW

You can put this directly into your lab record:

Airline Delay Communication Example

Aviation Data/Input

Flight and operational data



Traditional AI

Predicts whether a flight may be delayed



Prediction/Recommendation

Possible delay identified



Generative AI

Drafts and personalises passenger notification



AI-Generated Content

Passenger delay message



Human Verification

Authorised business-like verifies facts and content



Approved Business Action

Approved notification is communicated to passengers

This follows the final workflow structure specified in the lab.

FINAL REFLECTION

1. What is the main difference between Traditional AI and Generative AI?

Traditional AI primarily predicts, detects, classifies, optimises or recommends using data, whereas Generative AI creates new content such as text, images, summaries and explanations.

2. Give one aviation situation where both can work together.

An airline can use Traditional AI to **predict a flight delay** and Generative AI to **prepare a passenger notification** explaining the verified delay.

3. What was one limitation or mistake you observed in ChatGPT?

One limitation is that ChatGPT may provide details that sounds correct even when supporting information has not been provided. This is especially key when dealing with airline policies or regulations.

4. Give one example of an aviation hallucination.

Giving a specific **compensation amount or refund timeline for a cancelled flight without knowing the airline's policy or applicable regulation** would be an example of an aviation hallucination.

5. Which part of your workflow required human judgement?

The verification and approval of the AI prediction and passenger communication needed human judgement.

6. Why is human verification especially important in aviation?

Aviation involves passenger safety, operations, regulations and business-like responsibility. Incorrect AI-generated details can therefore have serious consequences.

7. How could this AI workflow improve passenger experience or airline/airport operations?

Traditional AI can identify potential delays early, while Generative AI can help prepare straightforward and personalised communication. This can help the organisation communicate details more efficiently while keeping human verification in the process.

8. Which aviation decisions should never be made solely on the basis of a Generative AI response?

Generative AI should not independently make **safety-critical, airworthiness, emergency, regulatory, legal or serious safety-investigation decisions**. These require authorised human professionals.

The lab's overall takeaway is: **AI prepares → Human verifies → Human decides**.

LAB RECORD — FINAL ORDER

For your submission, arrange everything in this order:

1. **Traditional AI vs Generative AI rules**
 2. **Initial classification of all 15 situations**
 3. **ChatGPT classification**
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4. **Comparison table**
5. **Selected case — Case A**
6. **Original combined workflow**
7. **Workflow-review checklist**
8. **Improved workflow**
9. **Completed 2×2 matrix**
10. **Hallucination challenge**
11. **Final aviation workflow**
12. **Final reflection**